

Santiago Páez Avendaño

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EDUCATION

M.Sc. student in Astrophysics

Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM)

Mexico City, Mexico

Aug. 2024 – Expected Aug. 2026

- Thesis: Data reduction method for OPTICAM multiband time series of transiting exoplanets
- Advisor: Yilen Gómez Maqueo Chew
- GPA: 9.44/10.0

B.A. in Natural Sciences Education

Faculty of Education, Universidad de La Sabana

Chia, Colombia

Jan. 2019 – Mar. 2024

- Awarded thesis: “Exoplanet Hackathon: A challenge-based learning proposal for the development of scientific skills”.
- Advisor: David Tovar
- GPA: 4.16/5.0

RESEARCH AND ACADEMIC EXPERIENCE

SCIENTIFIC ARTICLES

- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. (2026). “Data reduction method for OPTICAM multiband time series of transiting exoplanets”. *RAS Techniques and Instruments* 5, rzag021.
- Khandelwal, A. et al. [included **Páez, S.**] (2026). “TOI-6884: A Low-Mass Brown Dwarf Transiting an F-Type Star”. *Monthly Notices of the Royal Astronomy Society* In revision during May 2026.
- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] (2026). “TOI-1080 b: a temperate, rocky planet orbiting a quiet M4V host”. *Monthly Notices of the Royal Astronomical Society* 548.
- **Páez, S.**, García, I., and Acevedo, A. (2022). “Enseñanza de la expresión genética: modelo y modelización en grado décimo”. *Bio-grafía* 15(29), pp. 54–66.
- Leal Leal, M. et al. [included **Páez, S.**] (2021). “Mars Dichotomy: Prospects for human life on Mars”. *EarthArXiv eprints* X51S5J.

ACCEPTED OBSERVING PROPOSALS

- **Co-I.** Insights into stellar activity cycles: Spot crossing during transit. Time: 3 nights in Mar. 2025 and 3 nights in Mar. 2026 Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Data acquisition and data reduction.
- **Co-I.** Mass Measurements of Two Gas Giant Planets around M-dwarf stars. Time: 1.75 nights in 2026A. Instrument: WIYN/NEID. Role: Preparation of observing proposal and transit fitting.
- **Co-I.** Multi-band Transit Follow-up of Giant Planet Candidates Around M-Dwarfs with OPTICAM. Time: 6 nights in Feb. 2025 and 6 nights in Feb. 2026 Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Preparation of observing proposal, design of observation plan, data acquisition, data reduction, and transit fitting.

CONFERENCES, MEETINGS AND POSTERS

- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. Data reduction method for OPTICAM multiband time-series of transiting exoplanets. *Oral presentation in the XXII Astronomy Students Meeting (REA)*. Mexico City, Mexico. 2025.
- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. Método de reducción de datos para series de tiempo multibanda con OPTICAM de exoplanetas transitantes. *Seminar of Astrophysics and Cosmology COSMOS - Instituto de Ciencias Físicas - UNAM*. Online. 2025.

- **Páez, S.**, Tovar, D., and Barreto, C. *Proyéctate: a novel strategy to reach the public audience using Astronomy as a catching science topic*. Oral Presentation. *Communicating Astronomy with the Public - CAP*. 2024.
- León-García, A. et al. [included **Páez, S.**] Follow-up of WASP-43b and WASP-107b transits through differential photometry from OAN-TNT. Poster. *LXVI National Physics Congress*. Morelia, Mexico. 2023.
- **Páez, S.**, Puentes, G. A., and Becerra, D. *Arduino Uno Models for Teaching about Exoplanets in Transit and Radial Velocity*. Oral Presentation. *XI National Congress on Physics and Astronomy Teaching*. Medellín, Colombia. 2022.
- Tovar, D. et al. [included **Páez, S.**] *Astronomy Festivals in Cundinamarca (Colombia) as Educational Strategies for Learning Scientific Concepts*. Poster. *Communicating Astronomy with the Public - CAP*. 2021.

INTERNSHIPS, WORKSHOPS, AND SUMMER SCHOOLS

Code/Astro Workshop

Evanston, IL, USA

A week-long intensive training in astronomy software development. It covered the fundamental software engineering skills and best practices necessary for building sustainable open-source packages for astronomy applications, such as documentation, code profiling, unit tests, parallelization, code packaging and release.

Aug. 4–8, 2025.

XV ESAOBELA

Puebla, Mexico

100 hours of astronomical observation using 1-meter and 10-inch telescopes, as well as 51 hours of theoretical instruction at the Observatorio Astronómico Nacional de Tonantzintla (Puebla, México).

Jan. 2023 – Feb. 2023.

XXXI Scientific Summer at the OAN-SPM

Online

Summer school featuring theoretical classes on general astrophysics topics and remote observing sessions using the 84-cm telescope at the Observatorio Astronómico Nacional en la Sierra de San Pedro Mártir (Baja California, México).

Jun. 2022.

Alpha-Cen Summer School

Online

Summer School focused on developing programming skills for handling astronomical data and introducing core topics in astrophysics.

Aug. 2021.

Pedagogical/Teaching Internships

Chia, Bogota, Sopo, Zipaquirá, (Colombia)

Assisted in the instruction of Chemistry, Biology, and Physics for primary and secondary students across public and private institutions. Additionally, served as a museum guide and science communicator at interactive museums and the Bogota Planetarium.

Aug. 2020 – Nov. 2023.

RECOGNITIONS AND SCHOLARSHIPS

National Graduate Scholarship (Master's Degree)

Mexico City, Mexico

Recipient of fully the funded National Graduate Scholarship by the Secretariat of Science, Humanities, Technology and Innovation (SECIHTI) of Mexico

Aug. 2024 – Jun. 2026

Bogota City Council

Bogota, Colombia

Recognition through an official council note for being a Top 50 Finalist of the Global Student Prize 2023 (Proposition No. 521 of 2023).

Dec. 2023

Best Saber Pro test score

Chia, Colombia

Awarded for achieving the highest Saber Pro undergraduate examination results in the Natural Sciences Education program during 2022

Oct. 2023

Global Student Prize 2023

Varkey and Chegg.org Foundations

Top 50 Finalist. In recognition of exceptional achievements in the classroom and beyond. Only Colombian in the top 50 finalists out of 5,600 applicants worldwide.

Jul. 2023

FEST Scholarship

Fully funded undergraduate scholarship by Secretariat of Education of Bogotá

Bogota, Colombia
Jan. 2019 – Nov. 2024**OUTREACH**

Public Talk

“Ultra-precise light curves for studying transiting exoplanets. Clavius Astronomical Center of the Universidad Iberoamericana”.

Mexico City, Mexico
Feb. 2026**Staff Noche de Las Estrellas Festival**

Outreach staff in the Guillermo Haro Planetarium during the Noche de Las Estrellas Festival.

Mexico City, Mexico
Nov. 2025**Testimonial**

“In search of exoplanets. The transit of a Colombian astronomer in Mexico”. A testimonial describing the experience of pursuing a Master’s degree in astrophysics at UNAM.

Journal UNAM International
Mar. 2025**Public Talk**

“Mexico in the discovery and study of distant worlds”. Orizaba Planetarium.

Orizaba, Mexico
Feb. 2025**Astronomy Outreach Festivals**

Organized and participated in eight astronomy outreach festivals across multiple municipalities in the Sabana Centro province. Activities included workshops, public lectures, and nighttime observations, reaching over 4,000 participants in total.

Sabana Centro Province, Colombia
May 2019 – Nov. 2023**LANGUAGES**

- Spanish – Native
- English – TOEFL exam: 79 (Reading: 23, Listening: 13, Speaking: 23, Writing: 20)

SKILLS

Programming

Python Git/GitHub MATLAB HTML/CSS R Studio

SoftwareJupyter Notebook AstrolImageJ Poetry UV $\text{\LaTeX}$ Topcat VS Code**Scientific Packages**

Astropy photutils Scipy juliet exoplanet pyaneti PyMC

Operating Systems

macOS Linux